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Medical operation apparatus

Abstract

A medical operation apparatus includes a body graspable by an operator, an operation part that is arranged to enable a part of the operation part to be moved relative to the body, and a holder that receives the fingers of the operator and transmits a movement of the fingers to the operation part. The holder includes a contact portion that comes in contact with the body, and a connector that supports a relative posture of the holder and the operation part so as to change the relative posture in accordance with a posture change of the fingers of the operator. In a relative arrangement position of the operation part and the body, when the relative posture of the holder and the operation part is changed, an amount of change in the relative arrangement position of the body and the operation part is within a specified permissible range.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of International Patent Application No. PCT/JP2020/026766 filed on Jul. 8, 2020, in the Japan Patent Office, the contents of which being incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(1) The present disclosure relates to a medical operation apparatus that allows to reflect a grasping operation performed by an operator.

2. Description of Related Art

(2) In recent years, endoscopic surgery using a master-slave type surgical assist robot has become widely available. Here, the master means a medical operation apparatus that remotely controls a slave, and the slave means a surgical device that performs surgery in accordance with instructions from the master.

(3) The above-described surgical assist robot has a configuration in which in response to an operator moving a grip of the master, slave forceps are moved in conjunction with the grip. However, it is difficult operators with different finger sizes and finger lengths to properly reflect a content of the movement resulting in unintentional operation.

SUMMARY

(4) It is an aspect to provide a medical operation apparatus which may easily inhibit an unintentional operation from being reflected in the medical operation apparatus.

(5) According to an aspect of one or more embodiments, there is provided a medical operation apparatus comprising a body to be grasped by an operator; an operation part arranged to enable at least a part of the operation part to be moved, by the operator, closer to and away from the body; and a holder configured to hold fingers of the operator grasping the body, the holder being configured to transmit a movement of the fingers to the operation part. The holder comprises a contact portion configured to be in contact with the body when the operation part is moved closer to the body, and a connector supporting a relative posture of the holder and the operation part so as to change the relative posture in accordance with a posture change of the fingers of the operator, wherein, in a relative arrangement position of the operation part and the body with which the contact portion is in contact, when the relative posture of the holder and the operation part is changed, an amount of change in the relative arrangement position of the body and the operation part is within a specified permissible range.

(6) According to another aspect of one or more embodiments, there is provided a medical operation apparatus comprising a body; an operation part connected to the body such that at least a portion of the operation part is movable with respect to the body; and a finger holder configured to receive fingers of an operator who grasps the body, the finger holder configured to transmit a movement of the fingers to the operation part and comprising a contact portion configured to contact with the

body when the operation part is moved toward the body, and a connector rotatably supporting the contact portion, wherein, when the operation part contacts the body and a relative posture of the finger holder and the operation part is changed, an amount of change in a relative position of the body relative to the operation part is within a threshold range.

(7) According to yet another aspect of one or more embodiments, there is provided a medical operation apparatus comprising a body including a shaft and a cylindrical grip attached to one end of the shaft; an operating part movable relative to the body; a finger holder that is slidable along the operating part, the finger holder comprising a connector, and a contact portion that is rotatably connected to the connector and that contacts the shaft of the body when the operating part is moved relative to the body, wherein, when the contact portion contacts the shaft, the contact portion is rotatable such that a distance between a central axis of the connector and a contact point between the contact portion and the shaft remains constant.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and/or other aspects will become apparent and more readily appreciated from the following description of various embodiments, taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a block diagram showing one example of an overall configuration of a surgical assist robot, according to some embodiments;

(3) FIG. 2 is a perspective view showing an overview of a medical operation apparatus, according to some embodiments;

(4) FIG. 3 is another perspective view showing an overview of the medical operation apparatus, according to some embodiments;

(5) FIG. 4 is a plan view showing an overview of the medical operation apparatus, according to some embodiments;

(6) FIG. 5 shows an example of a holder, according to some embodiments;

(7) FIG. 6A and FIG. 6B show the holder before and after movement, according to some embodiments;

(8) FIG. 7A and FIG. 7B show, respectively, the medical operation apparatus according to some embodiments, and a medical operation apparatus in a modified example, according to some embodiments;

(9) FIG. 8A and FIG. 8B show a movement of a medical operation apparatus according to the related art as a comparative example;

(10) FIG. 9 shows an overview of a medical operation apparatus in a modified example, according to some embodiments; and

(11) FIG. 10 shows an overview of a medical operation apparatus in a modified example, according to some embodiments.

DETAILED DESCRIPTION

(12) As described above, a surgical assist robot may have a configuration in which in response to an operator moving a grip of the master, slave forceps are moved in conjunction with the grip. Therefore, the operator who operates the master-slave type surgical assist robot can operate the forceps of the surgical assist robot at will as if the forceps are part of him/herself. Specifically, a medical operation apparatus may allow an operator to hold a body so as to wrap the body with his/her palm, place his/her fingers onto an engagement portion of an operation part, and move his/her fingers in this state, thereby reflecting the content of the movement.

(13) However, in the above-described medical operation apparatus, a distance between the body and the engagement portion is fixed. Thus, it is difficult for operators with different finger sizes and

finger lengths to properly reflect the content of the movement.

(14) It is conceivable to provide an adjustment mechanism to adjust the distance between the body and the engagement portion arranged in the operation part. However, if the adjustment mechanism is moved by an unintentional operation by the operator and the operation part is moved in conjunction with the movement of the adjustment mechanism, the content of the movement is reflected based on the unintentional movement of the operation part. As a result, the slave may be remote-controlled based on the unintentional operation by the operator.

(15) In one aspect of the present disclosure, it is advantageous to easily inhibit the unintentional operation from being reflected in the medical operation apparatus that is the master.

(16) According to some embodiments, a medical operation apparatus may be configured to acquire an operation performed by an operator, the medical operation apparatus comprising a body, an operation part, and a holder. The operator grasps the body. The operation part is arranged so as to enable at least a part of the operation part to be moved closer to and away from the body by the operator. The holder is configured to hold fingers of the operator grasping the body, and the holder is configured to transmit a movement of the fingers to the operation part. The holder comprises a contact portion and a connector. The contact portion is configured to be in contact with the body when the operation part is moved closer to the body. The connector supports a relative posture of the holder and the operation part so as to change the relative posture in accordance with a posture change of the fingers of the operator. In the relative arrangement position of the body with which the contact portion is in contact and the operation part, when the relative posture of the holder and the operation part is changed, an amount of change in the relative arrangement position of the body and the operation part is within a specified permissible range.

(17) According to this configuration, the connector supports the relative posture of the holder and the operation part so as to change the relative posture in accordance with the posture change of the fingers of the operator. Therefore, the device is configured to be available for multiple operators with different finger sizes.

(18) Furthermore, in a state where the contact portion is in contact with the body, even if the operator performs an unintentional operation due to a force applied to the fingers and the posture of the operator's fingers and the relative posture of the holder and the operation part are changed, the amount of change in the relative arrangement position of the body and an operation part is within the specified permissible range.

(19) This configuration makes it possible to inhibit the operation part from moving equal to or more than the permissible range even if the unintentional operation is performed by the operator, and makes it easy to inhibit the operation part from moving equal to or more than the permissible range. As a result, in the medical operation apparatus that is the master, it is easier to inhibit the unintentional operation by the operator from being reflected.

(20) According to some embodiments, a sensor may be configured to detect the relative arrangement position of the operation part to the body. The permissible range may be less than a minimum value of an amount of change, detectable by the sensor, between the body and the operation part.

(21) In this configuration, the permissible range is set to less than the minimum value of the amount of change detectable by the sensor. Thus, it is possible to inhibit the operation part from moving equal to or more than the minimum amount of change detectable by the sensor even if the operator performs the unintentional operation. This configuration makes it easy to inhibit the detector from detecting the unintentional operation by the operator as a grasping operation.

(22) According to some embodiments, the contact portion may comprise a convex surface formed into a convex shape, and the amount of change in the relative arrangement position of the body and the operation part, based on a variation in a distance between the convex surface and a center of a posture change of the holder in the connector, may be within the specified permissible range.

(23) In this configuration, the amount of change in the relative arrangement position of the body

and the operation part, based on the variation in the distance between the convex surface of the contact portion and the center of the posture change of the holder in the connector, is within the specified permissible range. This configuration makes it is easy to inhibit the unintentional operation by the operator from being reflected as the grasping operation.

(24) According to some embodiments, the contact portion may comprise contact flat surfaces, and the amount of change in the relative arrangement position of the body and the operation part, based on a variation in a distance between each of the contact flat surfaces and a center of a posture change of the holder in the connector, may be within the specified permissible range.

(25) In this configuration, the amount of change in the relative arrangement position of the body and the operation part, based on the variation in the distance between each of the flat surfaces of the contact portion and the center of the posture change of the holder in the connector, is within the specified permissible range. This configuration makes it is easy to inhibit the unintentional operation by the operator from being reflected as the grasping operation.

(26) According to some embodiments, the body comprises a concave surface formed into a concave shape in an area to be in contact with the contact portion, and when the contact portion is in contact with the concave surface, the amount of change in the relative arrangement position of the body and the operation part, based on a variation in a distance between the concave surface and a center of a posture change of the holder in the connector, may be within the specified permissible range.

(27) In this configuration, when the contact portion is in contact with the concave surface formed into a curved surface in the body, the amount of change in the relative arrangement position of the body and the operation part, based on the variation in the distance between the concave surface and the center of the posture change of the holder in the connector, is within the specified permissible range. This configuration makes it is easy to inhibit the unintentional operation by the operator from being reflected as the grasping operation.

(28) According to some embodiments, the body may comprise body flat surfaces forming a concave shape in the area to be in contact with the contact portion, and when the contact portion is in contact with any of the body flat surfaces, the amount of change in the relative arrangement position of the body and the operation part, based on a variation in a distance between each of the body flat surfaces and a center of a posture change of the holder in the connector, may be within the specified permissible range.

(29) In this configuration, when the contact portion is in contact with the concave surface formed by the flat surfaces in the body, the amount of change in the relative arrangement position of the body and the operation part, based on the variation in the distance between each of the body flat surfaces and the center of the posture change of the holder in the connector, is within the specified permissible range. This configuration makes it is easy to inhibit the unintentional operation by the operator from being reflected as the grasping operation.

(30) According to some embodiments, the operation part may be formed into a long shape, and the connector connects the holder and the operation part so that an arrangement position of the holder along a longitudinal direction of the operation part having a long shape is changeable.

(31) In this configuration, the connector connects the holder and the operation part so that an arrangement position of the holder along the longitudinal direction of the operation part having a long shape is changeable. Therefore, the arrangement position of the holder can be varied relative to the operation part in accordance with the size or the like of the operator's fingers.

1. Configuration

(32) The surgical assist robot **1** may be a master-slave robot for endoscopic surgery and the like. The surgical assist robot **1** includes a master device and a slave device. The operation that has been reflected in the master device is transmitted to the slave device by communications, and the slave device works.

(33) The surgical assist robot **1** will be described with reference to FIG. **1**.

(34) The surgical assist robot **1** comprises a master device **100**, a controller **200**, and a slave device

300.

(35) A medical operation apparatus **101** is a part of the master device **100**, and is configured to detect a grasping operation performed by an operator, including a medical worker. The medical operation apparatus **101** is configured to be able to reflect at least the grasping operation performed by the operator. The detector **10A** is configured to detect the operation performed by the operator with a sensor arranged in a body, for example.

(36) The controller **200** is configured to control the slave device **300** based on the operation performed by the operator that has been detected by the medical operation apparatus **101**. The controller **200** comprises various control structures and control methods in accordance with the configuration of the slave device **300**.

(37) The slave device **300** is a slave-side device in the surgical assist robot **1**, and is configured to perform a grasping action based on the control by the controller **200**. For example, the slave device **300** may be a pair of forceps. Examples of a target grasped by the slave device **300** may include blood vessels and/or organs of a patient's body.

(38) The target grasped by the slave device **300** is not limited to the blood vessels or the organs of the patient's body, and may include various targets that can be grasped by the slave device **300**.

(39) Here, in addition to the grasping operation performed by the operator, the medical operation apparatus **101** may detect operations to move the slave device **300** and to change the posture of the slave device **300**. Thus, the controller **200** may control the movement and the posture change of the slave device **300** in addition to the grasping action. The medical operation apparatus **101** is configured to be operated by the operator's right hand; however, the medical operation apparatus **101** may be configured to be operated by the left hand.

(40) FIG. 2 is a perspective view showing an overview of a medical operation apparatus, according to some embodiments. FIG. 3 is another perspective view showing an overview of the medical operation apparatus, according to some embodiments. FIG. 4 is a plan view showing an overview of the medical operation apparatus, according to some embodiments. FIG. 5 shows an example of a holder, according to some embodiments.

(41) As shown in FIG. 1 to FIG. 5, the medical operation apparatus **101** according to some embodiments comprises a body **10**, an operation part **20**, a holder **30**, a rotating portion **40**, a detector **10A**, and a supporting portion **50**. The illustration of the structure in which the medical operation apparatus **101** is connected to the master device **100** is omitted in FIGS. 1-5.

(42) The body **10** is a part to be grasped by the operator. The body **10** comprises a long portion **12** formed into a long shape such as a columnar shape, and a grip **13** protruding from an end of the long portion **12**. In some embodiments, the long portion **12** may be a bar, a shaft, or a beam. In the body **10**, the grip **13** is formed at one end of the long shape. The body **10** is configured to be held by the operator with the grip **13** being grasped.

(43) Hereinafter, a longitudinal direction of the long portion **12** is assumed as a y-axis direction. The direction from the end with the grip **13** to the other end in the long portion **12** is assumed as a positive direction of the y-axis. An extending direction of the grip **13** perpendicular to the y-axis direction is assumed as a z-axis direction. The direction away from the long portion **12** is assumed as a positive direction of the z-axis direction. A direction perpendicular to each of the y-axis direction and the z-axis direction is assumed as an x-axis direction. The x-axis, the y-axis, and the z-axis are configured in accordance with a right-handed coordinate system.

(44) The grip **13** is formed into a substantially columnar shape. In some embodiments, the grip **13** may be, for example, a cylindrical shaft.

(45) The peripheral surface of the grip **13** may be formed into a shape easily held by the operator's right hand. Specifically, the peripheral surface of the grip **13** may have a concave shape suitable for the shape of the right hand.

(46) The grip **13** comprises a supporting portion **50** that is arranged near the long portion **12** and that has a long shape extending toward a substantially x-axis negative direction. The supporting

portion **50** comprises a first end that is on a grip **13** side and that is connected to the grip **13**. The supporting portion **50** comprises a second end that is opposite the first end and that comprises a rotating portion **40**.

(47) The operation part **20** is used to reflect the grasping operation performed by the operator. The operation part **20** has a long shape extending from the rotating portion **40**. In some embodiments, the operation part may be, for example, a beam, a rod, or a shaft.

(48) The rotating portion **40** is arranged between the supporting portion **50** and the operation part **20** and comprises a rotation axis extending in the substantially z-axis direction. The rotating portion **40** allows the supporting portion **50** and the operation part **20** to relatively move around the rotation axis. The rotating portion **40** rotates around the rotation axis, thereby enabling a change in the angle formed by a longitudinal direction of the supporting portion **50** and a longitudinal direction of the operation part **20**.

(49) In other words, when the operation part **20** rotates around the rotation axis of the rotating portion **40**, the operation part **20** moves around the rotation axis. Due to the rotation, the operation part **20** moves closer to or away from the body **10**.

(50) The detector **10A** is configured to detect the angle formed by the longitudinal direction of the supporting portion **50** and the longitudinal direction of the operation part **20**. The detector **10A** is configured to output, to the controller **200**, the detected angle as a detection signal. The detector **10A** may be arranged any position as long as the detector **10A** can detect the angle formed by the longitudinal direction of the supporting portion **50** and the longitudinal direction of the operation part **20**. The detector **10A** is an angle sensor for measuring the angle formed by the longitudinal direction of the supporting portion **50** and the longitudinal direction of the operation part **20**. The detector **10A** may be a rotary encoder or other angle sensors.

(51) The holder **30** is configured to hold fingers of the operator grasping the body **10**. The holder **30** is configured to transmit, to the operation part **20**, the movement of the fingers. The holder **30** is provided to the operation part **20**, and is configured such that the arrangement position thereof can be changed along the longitudinal direction of the operation part **20**. In other words, the holder **30** may slide along the operation part **20**. In operation, a hand of the operator may grasp the body **10** such that the operator's palm would contact the grip **13**, the operator's thumb would rest on the body **10** and the operator's fingers would wrap around and contact the holder **30**.

(52) The holder **30** comprises a connector **31**, a contact portion **33**, and a transmitting portion **35**.

(53) The connector **31** is connected to the operation part **20**. The connector **31** may be connected to the operation part **20** so that the holder **30** is movable along the longitudinal direction of the operation part **20**. In other words, the operation part **20** may slide along the beam, rod, or shaft.

(54) The connector **31** has a rotation axis along the substantially z-axis direction. The rotation axis allows the contact portion **33** and the transmitting portion **35** to rotate relative to the connector **31**.

(55) The connector **31** supports a relative posture between the contact portion **33**/the transmitting portion **35** and the operation part **20** so as to change the relative posture in accordance with the posture change of the fingers on the operation part **20**.

(56) When the operation part **20** comes closer to the long portion **12** of the body **10**, the contact portion **33** is in contact with a body right surface **11** of the long portion **12**. The body right surface **11** of the body **10** as used herein is the right surface of the body **10**, i.e. a side surface located on an x-axis negative side.

(57) Specifically, the contact portion **33** is arranged so as to be positioned on an x-axis positive side relative to the connector **31**. When positioned on the x-axis positive side, the contact portion **33** comprises a convex surface **33a** formed into a convex shape that is swelled in an x-axis positive direction. That is, the contact portion **33** comprises the convex surface **33a** outwardly convex relative to the connector **31** of the holder **30**.

(58) The convex surface **33a** is, for example, a circumferential surface with a rotation axis of the connector **31** as a central axis. That is, in a cross sectional view perpendicular to the central axis,

the central axis and each point on the convex surface **33a** are arranged so that a distance therebetween is substantially constant. Here, the arrangement is not limited to the one in which the distance between the central axis and each point on the convex surface **33a** is constant. Specifically, the convex surface **33a** may be formed so that a variation in the distance between the central axis and each point on the convex surface **33a** is within the permissible range **R**.

(59) Here, an example of the permissible range **R** may be a range in which the amount of change in angle between the long portion **12** of the body **10** and the operation part **20**, based on the variation in the distance between the central axis and each point on the convex surface **33a**, is less than a minimum value of the amount of change in angle detectable by the detector **10A**.

(60) The transmitting portion **35** is a flat portion to be in contact with the operator's finger, e.g. the index finger. The transmitting portion **35** is a surface opposite the contact portion **33** of the holder **30**. That is, in the holder **30**, the transmitting portion **35** is located opposite the contact portion **33** relative to the connector **31**.

2. Action

(61) FIGS. **6A-6B** show states where the holder **30** rotates around the rotation axis of the connector **31** while the contact portion **33** of the holder **30** is in contact with the body right surface **11** of the long portion **12** of the body **10**. For example, when the holder **30** rotates around the rotation axis of the connector **31**, the state thereof changes between a state shown in FIG. **6A** and a state shown in FIG. **6B**.

(62) In the state of FIG. **6A**, on a surface of the convex surface **33a**, a point in contact with the body right surface **11** is assumed as a first contact point **P1**, and a distance from the center of the rotation axis of the connector **31** to the first contact point **P1** is assumed as a first contact distance **D1**. When the holder **30** is rotated around the rotation axis of the connector **31**, in the state of FIG. **6B** and on the surface of the convex surface **33a**, a point in contact with the body right surface **11** is assumed as a second contact point **P2**, and a distance from the center of the rotation axis of the connector **31** to the second contact point **P2** is assumed as a second contact distance **D2**.

(63) Here, the convex surface **33a** is formed into a shape that is a circumferential surface around the rotation axis, and thus, the first contact distance **D1** and the second contact distance **D2** are the same. In other words, between the state shown in FIG. **6A** and the state shown in FIG. **6B**, a distance between the center of the rotation axis and the body **10** does not change even if the holder **30** is rotated around the rotation axis of the connector **31**. Therefore, a distance between the operation part **20** connected by the connector **31** and the body **10** also does not change due to the rotation.

(64) That is, the amount of change in the relative arrangement position of the body **10** and the operation part **20**, based on the variation in the distance between the convex surface **33a** and the central axis that is the center of the posture change of the holder **30** in the connector **31**, is within a specified permissible range **R**. In the relative arrangement position of the body **10** in contact with the portion **33** and the operation part **20**, when the relative posture of the holder **30** and the operation part **20** are changed, the amount of change in the relative arrangement position of the body **10** and the operation part **20** is within the specified permissible range **R**.

(65) The detector **10A** detects a change in the grasping operation based on the amount of change in the relative arrangement position of the body **10** and the operation part **20**. Thus, the detector **10A** does not detect a change in the grasping operation performed by the operator. As a result, there is no change in the control by the controller **200**, which performs the control based on the change in the grasping operation detected by the detector **10A**. Therefore, there is no change in the control of the slave device **300** in response to the rotation.

(66) That is, even if the holder **30** is rotated by an unintentional operation due to a force applied to the operator's finger touching the transmitting portion **35** in a state where the contact portion **33** of the holder **30** is in contact with the body **10**, it is easy to inhibit a change in the grasping by the slave device **300**.

3. Effects

(67) In the medical operation apparatus **101**, the connector **31** comprises the rotation axis, and the holder **30** rotates around the rotation axis, thereby making it possible to cope with the posture change of the fingers.

(68) This configuration allows the connector **31** to support the relative posture of the holder **30** and the operation part **20** so as to change the relative posture in accordance with the posture change of the operator's fingers.

(69) The connector **31** is arranged so as to be movable along the longitudinal direction of the operation part **20**, and thus adjustable for multiple operators with different finger sizes.

(70) Furthermore, in a state where the contact portion **33** is in contact with the body **10**, there is a possibility that the operator may perform an unintentional operation due to a force applied to the fingers, and the posture of the operator's fingers and the relative posture of the holder **30** and the operation part **20** may be changed. When the relative posture of the holder **30** and the operation part **20** is changed, the distance between the rotation axis of the connector **31** and the contact point is unchanged against the change in the position of the contact point. That is, the amount of change in the relative arrangement position of the body **10** and the operation part **20** is within the specified permissible range R.

(71) This configuration makes it possible to inhibit the operation part **20** from moving equal to or more than the permissible range R when the unintentional operation by the operator is performed, and makes it easy to inhibit the operation part **20** from moving equal to or more than the permissible range R.

(72) Specifically, FIG. 8 shows a comparative example, in which the holder **130** is formed into a rectangular parallelepiped shape and a body **110** comprises a flat side surface in a part to be in contact with a contact portion **133** of the holder **130**. In the case shown in FIG. 8, the holder **130** rotates around the connector **131** in a contact state as shown in FIG. 8A and FIG. 8B. Then, with the rotation of the holder **130**, a leading end of the operation part **120** is moved. When a sensor of a detector detects this movement, a controller controls the slave device **300** in accordance with the grasping operation detected by the detector.

(73) Thus, in the contact state, when the holder **130** is rotated around the connector **131** by the unintentional operation by the operator, the slave device **300** is controlled based on the unintentional operation, and the slave device **300** will perform an unintentional operation.

(74) The type of surgery for which the surgical assist robot **1** may be used is not particularly limited. However, when the surgical assist robot **1** is used for surgery requiring an operation to grasp targets that are different in hardness, such as blood vessels and organs, an unintentional operation is especially likely to occur. Therefore, the surgical assist robot **1** is suitable for use in surgery requiring an operation to grasp targets that are different in hardness.

(75) In some embodiments, even if the unintentional operation by the operator is reflected in the state where the holder **30** is in contact with the body **10**, the amount of change in the relative arrangement position of the body **10** and the operation part **20** is within the specified permissible range R.

(76) This configuration makes it possible to inhibit an unintentional grasping operation from being reflected in the detector **10A**. As a result, it is possible to easily inhibit the slave device **300** from performing an unintentional operation based on the unintentional grasping operation.

4. Modified Examples

(77) (1) In some embodiments, the convex surface **33a** forms the circumferential surface around the central axis of the connector **31**. However, the shape of the convex surface **33a** is not limited to the circumferential surface around the central axis of the connector **31**. For example, the connector **31** may be configured to have a specified center point and form a spherical shape around the center point like a sphere joint or a ball joint.

(78) Furthermore, in some embodiments, the distance from the central axis of the connector **31** to

the convex surface **33a** is constant in a cross section perpendicular to the central axis of the connector **31**. However, in some embodiments, the distance from the central axis of the connector **31** to the convex surface **33a** is not necessarily constant, and may be varied within the permissible range R. Here, in some embodiments, the permissible range R is exemplified as a range in which the amount of change in angle between the long portion **12** of the body **10** and the operation part **20**, based on the variation in the distance between the central axis and each point on the convex surface **33a**, is less than the minimum value of the amount of change in angle detectable by the detector **10A**. That is, the permissible range R is set to a value less than the minimum value of the amount of change detectable by the detector **10A**.

(79) Thus, it is possible to inhibit the operation part from moving equal to or more than the minimum value of the amount of change detectable by the detector **10A** in response to the unintentional operation by the operator. This configuration makes it easy to inhibit the detector **10A** from detecting the unintentional operation by the operator as the grasping operation.

(80) In other words, the amount of change in the relative arrangement position of the body **10** and the operation part **20**, based on the variation in the distance between the convex surface **33a** of the contact portion **33** and the center of the posture change of the holder **30** in the connector **31**, is within the specified permissible range R. This configuration makes it is easy to inhibit the unintentional operation by the operator from being reflected as the grasping operation.

(81) (2) In some embodiments, as shown in FIG. 7A, the contact portion **33** comprises the convex surface **33a** that is a circumferential surface. However, in some embodiments, the configuration is not limited to the one comprising the circumferential surface. As shown in FIG. 7B, the contact portion **33** may comprise holder-side contact flat surfaces **33b-33e**. The variation in a distance between each of the holder-side contact flat surfaces **33b-33e** and the central axis of the connector **31** that is the center of the posture change of the holder **30** in the connector **31** may be within the permissible range R.

(82) That is, based on the variation in the above distance, the variation in the distance between the central axis of the connector **31** and the body **10** may be configured to be within the permissible range R in a state where one of the holder-side contact flat surfaces **33b-33e** of the contact portion **33** is in contact with the body **10** and a in state where any other surface is in contact with the body **10**.

(83) With this configuration, the amount of change in the relative arrangement position of the body **10** and the operation part **20**, based on the variation in the distance between each of the holder-side contact flat surfaces **33b-33e** of the contact portion **33** and the center of the posture change of the holder **30** in the connector **31**, is within the specified permissible range R. This configuration makes it is easy to inhibit the unintentional operation by the operator from being reflected as the grasping operation.

(84) Note that the number of the holder-side contact flat surfaces **33b-33e** is not limited to four as shown in FIG. 7, and in some embodiments, the number may be four or more and four or less.

(85) (3) In some embodiments, a configuration has been explained in which the operator operates the device with the right hand. However, the medical operation apparatus **101** is not limited to the configuration applied to the right-handed operation. That is, in some embodiments, the device may be applicable to a left-handed operation. In this case, for example, each component constituting the medical operation apparatus **101** may be relatively arranged so as to be operated with the left hand. Specifically, the components of the medical operation apparatus **101** may be arranged to have a positional relationship that is mirrored relative to a y-z flat surface.

(86) (4) In some embodiments, the sensor of the detector **10A** is an angle sensor. However, in some embodiments, the sensor of the detector **10A** is not limited to the angle sensor. The sensor may be any sensor that can detect the arrangement position of the operation part **20** relative to the body **10**. For example, various kinds of sensors including a position sensor may be used.

(87) (5) In some embodiments, the contact portion **33** of the holder **30** comprises the convex

surface **33a**. This configuration makes it possible not to change the distance between the body **10** and the operation part **20** even if the holder **30** rotates around the central axis of the connector **31** in response to the posture change of the operator's fingers in a state where the convex surface **33a** is in contact with the body **10**. However, the configuration in which the distance between the body **10** and the operation part **20** does not change is not limited to the configuration in which the holder **30** comprises the convex surface **33a**.

(88) For example, as shown in FIG. **9**, in some embodiments, the body **60** may be formed to comprise, on a body right surface **61** of the body **60**, a concave surface **63a** concavely formed in an area to be in contact with the contact portion **33**. The shape of a contact portion **73** of a holder **70** may be formed into a shape corresponding to the concave surface **63a** of the body **60**.

(89) Specifically, the concave surface **63a** may be formed into a shape allowing the distance between a contact point and the rotation axis of the connector **71a** to be constant assuming that a point where the concave surface **63a** is in contact with the contact portion **73** as the contact point when the contact portion **73** of the holder **70** is in contact with the concave surface **63a**. The holder **70** may be formed into a shape, such as a rectangular parallelepiped shape, in which the contact portion **73** does not comprise the convex surface **33a**, for example.

(90) That is, the concave surface **63a** may be configured such that the amount of change in the relative arrangement position of the body **10** and the operation part **20**, based on the variation in the distance between the concave surface **63a** and the center of the posture change of the holder **30** in the connector **31**, is within the permissible range **R** when the contact portion **73** of the holder **70** is in contact with the concave surface **63a**.

(91) With this configuration, when the contact portion **33** is in contact with the concave surface **63a** formed into a curved surface in the body **10**, the amount of change in the relative arrangement position of the body **10** and the operation part **20**, based on the variation in the distance between the concave surface **63a** and the center of the posture change of the holder **30** in the connector **31**, is within the specified permissible range **R**.

(92) This configuration makes it is easy to inhibit the unintentional operation by the operator from being reflected as the grasping operation.

(93) The concave surface **63a** is not limited to a shape allowing a distance between the contact point and the rotation axis of the connector **71** to be constant assuming that the point where the concave surface **63a** is in contact with the contact portion **73** as the contact point when the contact portion **73** of the holder **70** is in contact with the concave surface **63a**. That is, the shape of the concave surface **63a** may be any shape if the variation in the distance between the contact point and the connector **71** of the rotation axis is within the permissible range **R**.

(94) (6) Furthermore, as shown in FIG. **10**, in some embodiments, the body **10** may comprise body contact flat surfaces **63b-63g** forming a concave shape in an area to be in contact with the contact portion **33**. When the contact portion **33** is in contact with any of the body contact flat surfaces **63b-63g**, the amount of change in the relative arrangement position of the body **10** and the operation part **20**, based on the variation in the distance between each of the body contact flat surfaces **63b-63g** and the center of the posture change of the holder **30** in the connector **31**, may be within the specified permissible range **R**.

(95) The distance between each of the body contact flat surface **63b-63g** and the center of the posture change of the holder **30** in the connector **31** as used herein may mean, for example, a length of a perpendicular line drawn from each of the body contact flat surfaces **63b-63g** towards the central axis of the connector **31**.

(96) In this configuration, when the contact portion **33** is in contact with the concave surface consisting of the body contact flat surfaces **63b-63g** of the body **10**, the amount of change in the relative arrangement position of the body **10** and the operation part **20**, based on the variation in the distance between each of the body contact flat surfaces **63b-63g** and the center of the posture change of the holder **30** in the connector **31**, is within the specified permissible range **R**. This

configuration makes it is easy to inhibit the unintentional operation by the operator from being reflected as the grasping operation.

(97) (7) Note that the number of the body contact flat surfaces **63b-63g** is not limited to six as shown in FIG. **10**, and in some embodiments, the number may be six or more and six or less.

(98) (8) A plurality of functions of one element of the aforementioned embodiments may be performed by a plurality of elements, and one function of one element may be performed by a plurality of elements. Furthermore, a plurality of functions of a plurality of elements may be performed by one element, and one function performed by a plurality of elements may be performed by one element. A part of the configurations of the aforementioned embodiments may be omitted. Furthermore, at least a part of the configurations of the aforementioned embodiments may be added to or replaced with the configurations of the other above-described embodiments.

(99) (9) Other than the medical operation apparatus **101** and the surgical assist robot **1**, the present disclosure may also be embodied in various forms, such as a system comprising the medical operation apparatus **101** and the surgical assist robot **1**, a program to functionalize a computer as the medical operation apparatus **101** and the surgical assist robot **1**, a non-transitory tangible recording medium such as a semiconductor memory storing the program, and a method for operating the medical operation apparatus **101** and the surgical assist robot **1**.

(100) It should be understood that embodiments are not limited to the various embodiments described above, but various other changes and modifications may be made therein without departing from the spirit and scope thereof as set forth in appended claims.

Claims

1. A medical operation apparatus comprising: a body to be grasped by an operator; an operation part arranged to enable at least a part of the operation part to be moved, by the operator, closer to and away from the body; and a holder configured to hold fingers of the operator grasping the body, the holder being configured to transmit a movement of the fingers to the operation part, wherein the holder comprises: a contact portion configured to be in contact with the body when the operation part is moved closer to the body, and a connector supporting a relative posture of the holder and the operation part so as to change the relative posture in accordance with a posture change of the fingers of the operator, and wherein, in a relative arrangement position of the operation part and the body with which the contact portion is in contact, when the relative posture of the holder and the operation part is changed, an amount of change in the relative arrangement position of the body and the operation part is within a specified permissible range.

2. The medical operation apparatus according to claim 1, further comprising a sensor configured to detect the relative arrangement position of the operation part relative to the body, wherein the specified permissible range is less than a minimum value of a detectable amount of change, detectable by the sensor, between the body and the operation part.

3. The medical operation apparatus according to claim 2, wherein the contact portion comprises a convex surface formed into a convex shape, and wherein the amount of change in the relative arrangement position of the body and the operation part, based on a variation in a distance between the convex surface and a center of a posture change of the holder in the connector, is within the specified permissible range.

4. The medical operation apparatus according to claim 2, wherein the contact portion comprises a plurality of contact flat surfaces, and wherein the amount of change in the relative arrangement position of the body and the operation part, based on a variation in a distance between each of the plurality of contact flat surfaces and a center of a posture change of the holder in the connector, is within the specified permissible range.

5. The medical operation apparatus according to claim 2, wherein the body comprises a concave surface formed into a concave shape in an area to be in contact with the contact portion, and

wherein, when the contact portion is in contact with the concave surface, the amount of change in the relative arrangement position of the body and the operation part, based on a variation in a distance between the concave surface and a center of a posture change of the holder in the connector, is within the specified permissible range.

6. The medical operation apparatus according to claim 2, wherein the body comprises a plurality of body flat surfaces forming a concave shape in an area to be in contact with the contact portion, when the contact portion is in contact with any of the plurality of body flat surfaces, the amount of change in the relative arrangement position of the body and the operation part, based on a variation in a distance between each of the plurality of body flat surfaces and a center of a posture change of the holder in the connector, is within the specified permissible range.

7. The medical operation apparatus according to claim 1, wherein the contact portion comprises a convex surface formed into a convex shape, and wherein the amount of change in the relative arrangement position of the body and the operation part, based on a variation in a distance between the convex surface and a center of a posture change of the holder in the connector, is within the specified permissible range.

8. The medical operation apparatus according to claim 1, wherein the contact portion comprises a plurality of contact flat surfaces, and wherein the amount of change in the relative arrangement position of the body and the operation part, based on a variation in a distance between each of the plurality of contact flat surfaces and a center of a posture change of the holder in the connector, is within the specified permissible range.

9. The medical operation apparatus according to claim 1, wherein the body comprises a concave surface formed into a concave shape in an area to be in contact with the contact portion, and wherein, when the contact portion is in contact with the concave surface, the amount of change in the relative arrangement position of the body and the operation part, based on a variation in a distance between the concave surface and a center of a posture change of the holder in the connector, is within the specified permissible range.

10. The medical operation apparatus according to claim 1, wherein the body comprises a plurality of body flat surfaces forming a concave shape in an area to be in contact with the contact portion, when the contact portion is in contact with any of the plurality of body flat surfaces, the amount of change in the relative arrangement position of the body and the operation part, based on a variation in a distance between each of the plurality of body flat surfaces and a center of a posture change of the holder in the connector, is within the specified permissible range.

11. The medical operation apparatus according to claim 1, wherein the operation part is formed into a long shape, and wherein the connector connects the holder and the operation part so that an arrangement position of the holder along a longitudinal direction of the operation part having a long shape is changeable.

12. A medical operation apparatus comprising: a body; an operation part connected to the body such that at least a portion of the operation part is movable with respect to the body; and a finger holder configured to receive fingers of an operator who grasps the body, the finger holder configured to transmit a movement of the fingers to the operation part and comprising: a contact portion configured to contact with the body when the operation part is moved toward the body, and a connector rotatably supporting the contact portion, wherein, when the operation part contacts the body and a relative posture of the finger holder and the operation part is changed, an amount of change in a relative position of the body relative to the operation part is within a threshold range.

13. The medical operation apparatus according to claim 12, further comprising a sensor configured to detect the relative position, wherein the threshold range is less than a minimum value of a detectable amount of change between the body and the operation part that is detectable by the sensor.

14. The medical operation apparatus according to claim 12, wherein the contact portion comprises a convex surface, and the amount of change is based on a variation in a distance between the

convex surface and a central axis of the connector.

15. The medical operation apparatus according to claim 12, wherein the contact portion comprises a plurality of contact flat surfaces, and the amount of change is based on a variation in a distance between each of the plurality of contact flat surfaces and a central axis of the connector.

16. The medical operation apparatus according to claim 12, wherein the body comprises a concave surface in an area to be in contact with the contact portion, and wherein the amount of change is on a variation in a distance between the concave surface and a central axis of the connector.

17. The medical operation apparatus according to claim 12, wherein the body comprises a plurality of body flat surfaces forming a contiguous concave surface in an area to be in contact with the contact portion, and the amount of change is based on a variation in a distance between each of the plurality of body flat surfaces and a central axis of the connector.

18. The medical operation apparatus according to claim 12, wherein the operation part comprises a shaft, and the connector connects the finger holder and the operation part so that an arrangement position of the finger holder along a longitudinal direction of the shaft is changeable.

19. A medical operation apparatus comprising: a body including a shaft and a cylindrical grip attached to one end of the shaft; an operating part movable relative to the body; a finger holder that is slidable along the operating part, the finger holder comprising a connector, and a contact portion that is rotatably connected to the connector and that contacts the shaft of the body when the operating part is moved relative to the body, wherein, when the contact portion contacts the shaft, the contact portion is rotatable such that a distance between a central axis of the connector and a contact point between the contact portion and the shaft remains constant.

20. The medical operation apparatus according to claim 19, wherein the contact portion has a convex surface that faces the shaft of the body.
